Asish Kumar Dalal

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Skills

Languages: JavaScript, C++17, Python, SQL
ML / AI: PyTorch, TensorFlow, Transformers (from scratch), MoE, GQA, Kernel Attention
Web: React, Vite, Node.js, Tailwind CSS, Web Crypto API, REST APIs
Systems & Tools: Linux (epoll, POSIX sockets), TCP/IP, Git, Docker

Experience

EnvSecure — Zero-Knowledge `.env` Sharing Tool — live ↗ — 2025 – Present

Creator

- Web + CLI tool for securely sharing `.env` files. Encryption (**AES-256-GCM**, PBKDF2 key) runs entirely in-browser — server never sees plaintext. Self-destructing links. **200+ organic users.** *React, Web Crypto API*

KeystoneJS (9.9k ★) — Open-Source Contributor — May 2025

PR #9830 merged by core maintainer dcousens

- Fixed three bugs in the `usecase-blog` example that blocked every new developer from running the project: added missing `status` and `publishedAt` fields to the **Prisma** schema, converted blog content from plain strings to the **Slate.js Document AST** format Keystone expects, and handled a null-author crash.

Projects

HSKM Architecture — O(N) Language Model

- Standard Transformers get slower as input grows ($O(N^2)$). Built a new attention design that runs in **O(N)** — each token looks up 64 learned memory slots using **Top-K sparse selection** instead of comparing with every other token.
- Added three memory layers: short-term (**sparse attention**), medium-term (**parallel EMA scan**, no slow recurrence), and long-term (4096 global slots with **gated read/write**). A learned gate fuses all three.
- Result: **4× less GPU memory** — ~6 GB at 8k context vs ~24 GB for a same-size Transformer. *Python, PyTorch*

C++ HTTP Load Balancer — Built from Scratch, No Libraries

- A complete TCP/HTTP load balancer in 17 KB of C++17 with zero external libraries. Uses Linux **epoll** for non-blocking I/O so it handles many connections without threads per client.
- Connections follow a 4-state lifecycle with **64 KB ring buffers**; data is fully drained before any socket closes. A background thread health-checks each backend every 5 s and removes dead servers from the pool. *C++17, Linux epoll, POSIX*

GPT-OSS (3.66B) — Language Model from Scratch

- Built a 3.66B-parameter language model from scratch in PyTorch (no HuggingFace). Uses **Mixture of Experts (MoE)** — a router picks the best expert layers for each token, saving compute. A load-balance loss keeps work spread evenly.
- Uses **Grouped Query Attention (GQA)** to cut KV cache memory by sharing key-value heads across queries. Built a unified **KV cache** with three modes (train / prefill / decode) so text generation never re-processes old tokens.
- Trained on 69M tokens on an **NVIDIA H200**; reached perplexity ≈ 3. *Python, PyTorch*

Education

B.Tech, Computer Science & Engineering — Budge Budge Institute of Technology — 2023 – 2027
Relevant Coursework: Deep Learning, Computer Vision, NLP, Algorithms, Operating Systems, Networks

Higher Secondary (Class XII) — Uluberia High School — 83.4% | 2023

Secondary (Class X) — Uluberia High School — 91.14%